Naveen Saragadam

Senior GenAI Engineer | Senior Data Engineer | RAG | LLM | Python | SQL
Email: naveens0519@gmail.com | **Phone:** +1 (480) 869-6966 | **LinkedIn:** linkedin.com/in/naveen-saragadam-de

Summary

- Senior GenAI / LLM Engineer with 10+ years of enterprise engineering experience across financial services, insurance, telecommunications, and healthcare, designing, developing, evaluating, and maintaining production-grade Generative AI applications, enterprise Retrieval-Augmented Generation (RAG) systems, and LLM-powered services. Build enterprise knowledge retrieval solutions that enable internal business teams, including operations, risk, and compliance, to access trusted, source-grounded information from enterprise documents and knowledge repositories.
- Design end-to-end RAG pipelines using Python, LangChain, LlamaIndex, and the OpenAI GPT-4 API, covering document ingestion, parsing, normalization, chunking, metadata enrichment, embedding generation, vector indexing, retrieval, reranking, and grounded response generation across FAISS, Pinecone, Qdrant, and pgvector. Evaluate LLM and RAG applications using representative evaluation datasets, expected answers, source documents, RAGAS, human assessment, citation validation, and automated regression testing, measuring retrieval relevance, Recall@K, answer correctness, faithfulness, and citation accuracy. Design controlled agentic workflows using function and tool calling, task decomposition, structured outputs, bounded execution, and human-in-the-loop approval.
- Also experienced in building and operationalizing machine learning pipelines using Azure Data Factory, Azure Databricks, and MLflow, supporting experiment tracking, feature engineering, model training, evaluation, and model lifecycle management, with hands-on MLOps experience in batch scoring, API-based inference, Docker-based packaging, Azure DevOps CI/CD, controlled model promotion across environments, scheduled model retraining, and production monitoring of prediction quality, latency, and data distribution changes.
- Experienced in production troubleshooting, incident analysis, root-cause investigation, release validation, and performance optimization across LLM applications, retrieval services, and supporting data pipelines, optimizing retrieval latency, application response times, caching efficiency, token utilization, and inference costs while maintaining answer quality and system reliability. Strong background in enterprise data engineering — ETL/ELT development, incremental ingestion, data quality, distributed processing, and ML-ready dataset preparation using Python, SQL, PySpark, Apache Spark, and Snowflake across Azure and AWS data services — delivering LLM-powered services with FastAPI and Docker under enterprise security and governance practices, including role-based access control, secrets management, audit logging, and document-level access restrictions.
- Experienced in enterprise data pipeline development using AWS Glue, Amazon S3, Amazon EMR, PySpark, and the AWS Glue Data Catalog, including telecommunications data cleansing, schema validation, standardization, deduplication, transformation, and production troubleshooting. Experienced in Snowflake ingestion and change-processing workflows involving Snowpipe, Streams, and Tasks, supporting incremental processing, pipeline dependencies, data validation, and downstream data availability. Familiar with dbt's role in SQL based transformation and analytics engineering, including the concepts of modular models, staging and intermediate layers, analytical marts, incremental processing, data tests, documentation, and dependency management. Existing hands-on experience is strongest in SQL, Python, PySpark, Delta Lake MERGE operations, enterprise data quality, and incremental pipeline development; hands-on dbt project configuration, model execution, testing, and deployment remain development areas rather than established production experience.

Skills

Programming Languages: Python, SQL, PySpark, Scala, Java, Bash, Shell Scripting, PL/SQL

Generative AI and LLMs: Large Language Models (LLMs), OpenAI GPT-4 API, Prompt Engineering, Retrieval-Augmented Generation (RAG), Enterprise Knowledge Retrieval, Document Question Answering, Grounded Generation, Structured Outputs, Response Validation, Hallucination Mitigation, Appropriate Abstention

LLM Frameworks and Orchestration: LangChain, LlamaIndex, RAG Orchestration, Prompt Templates, LLM Chain Development, Direct LLM API Integration, Context Assembly

RAG Pipeline Engineering: Document Ingestion, PDF Processing, Document Parsing, Text Normalization, Metadata Enrichment, Chunking Strategies, Embedding Generation, Vector Indexing, Incremental Ingestion, Reindexing, Semantic Retrieval, Hybrid Retrieval, Metadata Filtering, Top-K Retrieval, Reranking, Source Attribution, Citation Generation

Vector Databases and Search: FAISS, Pinecone, Qdrant, pgvector, Vector Embeddings, Similarity Search, Retrieval-Depth Tuning, Candidate Selection, Relevance Ranking, Retrieval Recall Optimization

Agentic AI: AI Agents, Function Calling, Tool Integration, Task Decomposition, Multi-Step Workflow Orchestration, State Management, Bounded Execution, Human-in-the-Loop Approval

LLM Evaluation and Quality Assurance: RAGAS, Human Evaluation, Representative Evaluation Datasets, Expected-Answer Comparison, Retrieval Relevance, Recall@K, Answer Correctness, Faithfulness, Citation Accuracy, Response Consistency, Abstention Evaluation, Citation Validation, BLEU, ROUGE, Regression Testing, Failure Analysis

MLOps & ML Engineering: MLflow, Experiment Tracking, Parameter Tracking, Model Evaluation Metrics, Experiment Reproducibility, Model Registry, Model Versioning, Model Artifact Management, Model Promotion, Model Lineage, Model Governance, Azure Databricks Jobs, Databricks Workflows, Feature Engineering Pipelines, Model Training Pipelines, Model Evaluation Pipelines, Automated Retraining, Batch Inference, Batch Scoring, REST API-Based Inference, Model Serving, Docker, ML CI/CD, Precision, Recall, F1-Score, ROC-AUC, RMSE, MAE, Prediction Quality Monitoring, Model Performance Monitoring, Data Drift Monitoring, Feature Consistency Validation

API Development, Deployment & Production Operations: FastAPI, REST APIs, Containerized Applications, Request Validation, Exception Handling, Configuration Management, Structured Logging, Request Tracing, Application Monitoring, Latency and Throughput Monitoring, p95 Response Time, Token Usage Monitoring, Inference Cost Optimization, Incident Analysis, Root-Cause Investigation, Production Troubleshooting, Operational Recovery

Security and Governance: Role-Based Access Control (RBAC), Secrets Management, Audit Logging, Document-Level Access Controls, Sensitive-Data Protection, Controlled Model Access, Enterprise Data Governance, Access Permissions

Data Engineering and ETL/ELT: Data Ingestion, Incremental Processing, Schema Validation, Data Quality, Deduplication, Data Standardization, Source-to-Target Reconciliation, Data Lineage, Reusable SQL Transformations, Business-Rule Validation, Pipeline Dependency Management, Failure Handling, Retries, Recovery Processing, Late-Arriving Data Handling

AWS Data Engineering: AWS Glue, AWS Glue ETL Jobs, AWS Glue Data Catalog, Amazon S3, Amazon EMR, AWS IAM, Batch Processing, CSV, JSON, Parquet, Metadata Management, Data Lake Architecture, Partitioned Data Storage

Azure Data Engineering: Azure Data Factory, Azure Databricks, ADLS Gen2, Azure SQL, Azure Synapse Analytics, Azure Key Vault, Delta Lake, Delta Lake MERGE, Auto Loader, Unity Catalog, Medallion Architecture (Bronze / Silver / Gold), Lakehouse Architecture

Snowflake, Warehousing & Databases: Snowflake, Snowpipe, Snowflake Streams, Snowflake Tasks, SQL Transformations, Data Availability Monitoring, Curated Analytical Datasets, SQL Server, PostgreSQL, Oracle, MySQL, Amazon Redshift, MongoDB, DynamoDB, Azure Cosmos DB

dbt and Analytics Engineering Concepts: dbt Fundamentals, SQL Model Design, Staging / Intermediate / Mart Architecture, Model Dependencies, ref() and source() Concepts, View and Table Materialization Concepts, Incremental Modeling Concepts, Data Testing Concepts, Source Freshness Concepts, Dimensional Modeling Concepts, Model Documentation, Git-Based Development, CI/CD Concepts

SQL and Distributed Processing: Spark SQL, HiveQL, Apache Spark, Spark UI, MERGE Operations, Joins, Aggregations, Query and Execution-Plan Analysis, Partition Optimization, Partition Pruning, Adaptive Query Execution (AQE), Broadcast Joins, Shuffle Optimization, Caching, Distributed Processing

Orchestration and Streaming: Apache Airflow, Oozie, Apache Kafka, Apache NiFi, Sqoop, Flume, Pig, Workflow Scheduling, Job Monitoring

Version Control and CI/CD: Git, Azure DevOps, CI/CD Pipelines, Automated Validation, Release Approvals, Environment-Specific Configuration, Controlled Deployment, Smoke Testing, Rollback Planning

Work Experience

JPMorganChase — Feb 2024 – Present

Senior GenAI Engineer | Contract — **TX**

- Design, develop, and support an enterprise Retrieval-Augmented Generation (RAG) platform enabling internal business teams, including operations, risk, and compliance, to retrieve relevant, source-grounded information from enterprise knowledge repositories, indexing approximately 250,000 documents and processing 5,000 queries per day.
- Achieve 92% answer correctness, 95% citation accuracy, and 90% Recall@5 on a curated evaluation dataset of 1,000 representative questions; reduce average response latency by approximately 30% through retrieval tuning, caching, and application-level performance improvements, and reduce LLM cost per query by approximately 20% through prompt optimization, more efficient context construction, and improved model usage.
- Develop modular RAG workflows using Python, LlamaIndex, and LangChain, integrating language models, embedding services, vector search, and retrieval components.
- Build document ingestion workflows for enterprise PDFs, internal policies, standard operating procedures, and knowledge articles, covering parsing, text normalization, metadata enrichment, chunking, embedding generation, and vector indexing; support document refresh, incremental ingestion, and reindexing workflows to keep the retrieval index synchronized with source repositories.
- Implement embedding-based semantic retrieval using FAISS, Pinecone, Qdrant, and pgvector, tuning similarity search parameters, retrieval depth, metadata filters, and candidate selection; optimize chunking, embedding and retrieval configurations, reranking, and prompt construction to improve answer relevance, source grounding, and response consistency.
- Evaluate LLM and RAG applications using RAGAS and complementary techniques, human assessment, citation validation, and automated regression testing, using BLEU and ROUGE where suitable for text-overlap evaluation, measuring retrieval relevance, answer correctness, faithfulness, citation accuracy, response completeness, and appropriate abstention for unsupported questions; investigate retrieval failures caused by missing documents, ineffective chunking, metadata inconsistencies, and embedding mismatches.
- Develop and maintain Python application services and REST APIs using FastAPI to expose retrieval and LLM capabilities to consuming applications, integrating embedding services, vector indexes, language

model APIs, and enterprise data sources with request validation, exception handling, and structured response handling.

- Implement controlled agentic tool-calling patterns using validated arguments, explicit execution boundaries, exception handling, and human-in-the-loop approval, keeping LLM-generated requests separate from application-side execution.
- Support production deployment, monitoring, and troubleshooting of GenAI applications using containerized services with Docker, monitoring request latency, p95 response time, throughput, error rates, token usage, and model inference cost; support release validation, smoke testing, regression testing, and rollback planning.
- Apply enterprise security controls including role-based access control, secrets management, audit logging, and document-level access considerations, collaborating with platform engineering, application development, information security, risk, and compliance stakeholders.
- Develop and maintain approximately 15–20 enterprise data pipelines processing approximately 50–100 GB per day, building reusable Python and SQL validation, reconciliation, and data-preparation components on Azure Data Factory, Azure Databricks, PySpark, and Spark SQL. Ingest new files incrementally with Auto Loader into a bronze layer, standardize into silver, and publish business-ready gold datasets in a medallion architecture. Apply Delta Lake MERGE operations for incremental updates and govern catalogs, schemas, and tables through Unity Catalog.
- Work with Snowflake ingestion and change-processing patterns, including Snowpipe, Streams, and Tasks, to support data availability and incremental processing, and support Apache Spark and Delta Lake workflows for data transformation and large-scale data preparation.

Environment: Python, SQL, FastAPI, REST APIs, LangChain, LlamaIndex, OpenAI GPT-4 API, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Semantic Search, FAISS, Pinecone, Qdrant, pgvector, Vector Indexing, Metadata Filtering, Hybrid Retrieval, Reranking, RAGAS, BLEU, ROUGE, Human Evaluation, Citation Validation, Regression Testing, Docker, Git, CI/CD, Structured Logging, Request Tracing, Caching, Snowflake, Snowpipe, Streams, Tasks, Apache Spark, PySpark, Delta Lake, Auto Loader, Unity Catalog, MERGE, Medallion Architecture, Azure Data Factory, Azure Databricks, ADLS Gen2, Azure Key Vault, RBAC, Secrets Management, Audit Logging

GEICO **Oct 2021 – Jan 2024**

Senior Data Engineer — ML & Data Analytics | Contract **Remote, TX**

- Developed and maintained approximately 20–30 enterprise data pipelines processing approximately 100–300 GB daily using Python, SQL, PySpark, Apache Spark, Azure Data Factory, and Azure Databricks, supporting insurance analytics and ML-ready dataset preparation; integrated SQL Server, Oracle, API-based sources, and flat-file feeds containing claims, policy, customer, and operational data.
- Built incremental ingestion and medallion-layer transformation workflows using Azure Data Factory, ADLS Gen2, Auto Loader, and PySpark, standardizing claims, policy, and customer data for downstream analytics and machine learning workflows; developed reusable SQL and Python transformation components for data cleansing, standardization, validation, deduplication, aggregation, and reconciliation, supporting consistent processing across recurring data pipelines and downstream analytical datasets.
- Implemented data-quality controls for required fields, duplicate records, schema consistency, business rules, referential integrity, record counts, and source-to-target comparisons, including schema validation and duplicate detection; investigated failed checks, corrected transformation or source-data issues, and validated recovery runs.
- Implemented incremental processing using business keys and Delta Lake MERGE operations to incorporate new records and update existing records without requiring unnecessary full reloads, applying deduplication and validation logic before merging records to maintain consistency in curated outputs; supported late-arriving data and historical corrections through controlled reprocessing, pipeline reruns, and

source-to-target reconciliation, investigating discrepancies and validating recovered datasets before downstream consumption.

- Developed curated analytical datasets from claims, policy, customer, and operational data for analytics and machine learning consumption, applying structured transformation stages and data validation to improve consistency and maintainability across downstream data workflows.
- Optimized a Spark workload by analyzing Spark UI metrics and physical execution plans, refining join strategies, filtering unnecessary data, and tuning partitions, reducing runtime from approximately 120 minutes to 90 minutes — a 25% improvement.
- Collaborated with data scientists to track experiments, parameters, datasets, and evaluation metrics using MLflow and Databricks, supporting model comparisons, experiment reproducibility, and model version management.
- Developed and maintained ML workflows using Azure Data Factory and Databricks Jobs to automate feature engineering, model training, evaluation, and data refreshes, incorporating dependency management, retries, and failure handling.
- Supported model deployment for batch scoring and API-based inference, troubleshooting prediction failures, dependency issues, and production performance; used Docker to package model-serving applications and maintain consistent execution environments.
- Implemented ML CI/CD workflows using Azure DevOps and Git to automate testing and deployment of ML code and pipelines, supporting release approvals, environment-specific configuration, and controlled promotion of model versions.
- Supported scheduled model retraining and production monitoring, evaluating refreshed models against predefined performance criteria and monitoring prediction quality, latency, job failures, and changes in production data distributions to identify potential degradation and retraining needs.
- Supported secure data processing and deployment practices using Azure Key Vault, Unity Catalog, role-based access controls, Git, and Azure DevOps workflows to manage credentials, permissions, and controlled releases.

Environment: Python, SQL, PySpark, Apache Spark, Spark SQL, Azure Data Factory, Azure Databricks, ADLS Gen2, Delta Lake, Auto Loader, Unity Catalog, MLflow, Databricks Jobs, Databricks Workflows, Azure DevOps, Git, Docker, CI/CD, Model Training, Experiment Tracking, Model Registry, Batch Scoring, REST APIs, Model Evaluation, Model Monitoring, Data Drift Monitoring, Azure SQL, Azure Key Vault, SQL Server, Oracle, Hadoop, Hive, MapReduce, Azure Synapse Analytics, Power BI

AT&T **May 2018 – Sep 2021**

Senior Data Engineer | Remote **Dallas, TX**

- Developed and maintained ETL pipelines using Python, SQL, PySpark, and Apache Spark to ingest, transform, validate, and publish telecommunications datasets — subscriber, customer usage, network operations, and billing records — for downstream analytics, reporting, and business intelligence.
- Built data ingestion workflows using Amazon S3 as a centralized data lake; developed and maintained AWS Glue ETL jobs using Python and PySpark, applying data cleansing, schema validation, standardization, deduplication, and transformation logic to telecommunications datasets.
- Used the AWS Glue Data Catalog to organize dataset metadata, support data discovery, and maintain metadata for data-processing workflows; worked with Glue ETL jobs and scheduled data-processing workflows, supporting recurring batch processing, pipeline dependencies, and production troubleshooting.
- Used Amazon EMR to run distributed Apache Spark and PySpark workloads; optimized Spark workloads by analyzing execution plans and job metrics, improving join strategies, filtering unnecessary records early, selecting required columns, tuning partitioning to reduce excessive data shuffling, and evaluating broadcast joins and caching strategies.

- Processed CSV, JSON, and Parquet datasets, implementing schema validation, null handling, duplicate detection, record-count comparisons, and source-to-target reconciliation; developed reusable Python utilities, SQL transformations, and PySpark jobs for cleansing, deduplication, aggregation, and business-rule validation across recurring batch workflows, organizing logic into reusable transformation stages that separate ingestion, cleansing, standardization, validation, and downstream data preparation, and applied consistent data-quality checks and reconciliation to help maintain reliable curated datasets.
- Troubleshot production ETL failures by reviewing execution logs, investigating transformation errors, validating source availability, and supporting controlled reruns and post-recovery data reconciliation; followed enterprise security and governance procedures, including least-privilege AWS IAM permissions and controlled access to Amazon S3 resources.
- Also built ingestion workflows using Sqoop, Flume, and Pig into HDFS and Hive, maintained Hive external tables, dynamic partitions, and bucketing strategies through Oozie workflows, developed Azure Data Factory pipelines between REST APIs, Blob Storage, Azure SQL, and analytical platforms, supported Kafka and Apache NiFi ingestion and streaming workflows, and modernized selected MapReduce workloads using Spark and Scala.

Environment: Python, SQL, PySpark, Apache Spark, Spark SQL, AWS Glue, AWS Glue ETL Jobs, AWS Glue Data Catalog, Amazon EMR, Amazon S3, AWS IAM, Hive, HiveQL, Hadoop, CSV, JSON, Parquet, ETL/ELT, Batch Processing, Incremental Data Processing, Data Quality Validation, Distributed Processing, Spark Execution-Plan Analysis, Broadcast Joins, Partition Optimization, Query Optimization, Git, CI/CD, Apache Airflow, Oozie, Sqoop, Flume, Pig, Apache Kafka, Apache NiFi, Azure Data Factory, Azure SQL, SSIS, REST APIs

UnitedHealth Group — Jan 2016 – Apr 2018

Data Engineer | On-site — **San Francisco, CA**

- Developed Hadoop-based ETL pipelines using Python, Sqoop, Hive, Spark, PySpark, and Oozie, ingesting Oracle and SQL Server data into Hadoop; developed Hive transformations and Spark SQL applications for cleansing, aggregation, and analytical consumption of JSON, Avro, and Parquet datasets.
- Built Kafka ingestion workflows and Oozie / Airflow orchestration for scheduled processing, implemented data-quality validation, reconciliation, and exception handling across recurring ETL workflows, and automated Amazon EMR cluster launches and Hadoop application configuration using Python scripts to standardize environment setup.
- Developed transformation workflows that separated ingestion, cleansing, validation, and analytical preparation, applying schema checks and reconciliation to improve the reliability of downstream datasets, and supported scheduled processing, dependency management, troubleshooting, and controlled reruns to recover failed workflows and validate downstream data availability.

Environment: Hadoop, HDFS, Hive, HiveQL, Sqoop, Spark, PySpark, YARN, Kafka, Oozie, Airflow, Amazon EMR, Python, Java, Shell Scripting, Avro, Parquet, JSON, Oracle, SQL Server

NCS Pearson India Private Limited — Mar 2013 – Apr 2014

Hadoop Developer Intern | On-site — **Delhi, India**

- Supported distributed Hadoop environments for big-data proof-of-concept initiatives, including multi-node cluster installation, configuration, balancing, and troubleshooting; developed Hive workflows, MapReduce programs, and Pig scripts and UDFs for distributed transformation, batch processing, and preprocessing.

Environment: Linux, SQL Server, Oracle, MongoDB, Hadoop, HDFS, Hive, MapReduce, Pig, HBase, Sqoop, Flume, ZooKeeper, Oozie, Python, Java, PL/SQL, Shell Scripting, Cloudera Manager, HDInsight

Education

University of Arizona — **Dec 2015**
Master of Science in Computer Science — **Tucson, AZ**

KMITE College, India — **May 2014**
Bachelor of Technology in Computer Science

Certifications

- AWS Certified Data Engineer — Associate
- Microsoft Certified: Azure Databricks Data Engineer Associate
- Microsoft Certified: Fabric Data Engineer Associate
- Databricks Certified Data Engineer Associate